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Measuring the Efficacy of Percutaneous Cryoneurolysis in Participants With Refractory or Plateaued Shoulder, Elbow, Wrist, or Finger Spasticity

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Objective: The aim of the study was to investigate novel, minimally invasive cryoneurolysis for refractory or plateaued upper limb spasticity.
Design: This was a repeated-measures, single-center, observational pilot study (NCT04670783). Percutaneous cryoneurolysis was applied to the upper limb nerves and intramuscular branches of 59 adults with upper limb spasticity refractory to or plateaued on botulinum therapies. Maximal passive range of motion (V1), active ROM, and modified Ashworth scale score were measured during shoulder flexion, abduction, and external rotation and elbow and wrist extension at baseline and follow-up to 1 yr. Additional outcomes included pain, participant satisfaction, and upper limb disability.
Results: Overall, 59 participants received cryoneurolysis in ≥1 region targeting nerves that innervate muscles supporting shoulder (n = 47), elbow (n = 33), wrist (n = 18), and fingers/thumb (n = 29) movement. At 12 mos, there was significant change from baseline in V1, active ROM, and modified Ashworth scale score for shoulder flexion and abduction and in V1 and modified Ashworth scale score for external rotation. Similar results were observed for elbow extension V1, active ROM, and modified Ashworth scale score and wrist extension modified Ashworth scale score. Average daily pain, participant satisfaction, and upper limb disability improved.
Conclusions: Participants with plateaued or refractory spasticity had improvements in upper limb regions; future evaluations of cryoneurolysis treatment for spasticity are warranted.

Key Words: Observational Study, Cryotherapy, Muscle Spasticity

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What Is Known

- Cryoneurolysis uses cold temperatures to cause reversible, second-degree nerve injury that can treat nerve pain. This treatment has been utilized for spasticity, but sample sizes and follow-up duration have been limited.

What Is New

- Cryoneurolysis can provide long-lasting improvements in upper limb spasticity, function, and pain, as well as participant satisfaction for up to 1 yr for patients with treatment-resistant spasticity as found in this repeated-measures, observational pilot study. Findings from this study support the potential use of cryoneurolysis to treat upper limb spasticity and provide a foundation for future research investigating cryoneurolysis for spasticity.

Spasticity arises from central nervous system injury. When severe, it leads to upper limb deformities that may have both a static or dynamic component.[1] The spastic upper limb is commonly held in an adducted and internally rotated position at the shoulder with flexion of the elbow and wrist and fingers.[2] Patients with severe spasticity receive treatments including oral antispasmodic medications, intrathecal baclofen pumps, botulinum toxins, alcohol, or phenol chemodenervation.[3–5] Severe spasticity may require surgical interventions such as tendon lengthening or neurectomy.[1] Disabling spasticity affects health, independence, and quality of life, and optimal outcomes may be unachievable with medications and botulinum toxin.[3,4,6,7] When surveyed, physicians in Canada, France, Germany, and the United States indicated that they were limited by maximum botulinum toxin dose restrictions allowed in their country based

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on regional indications and guidelines and that ~25% of their patients could benefit from higher doses.[3] Because severe spasticity results in pain affecting health and quality of life, there is a need for novel treatments to improve patient outcomes.[7,8]

Traditionally used in pain management,[9] percutaneous cryoneurolysis is a minimally invasive, ultrasound-guided procedure recently utilized as a novel treatment for spasticity.[8,10–12] A cryoprobe needle is inserted percutaneously to target a mixed motor sensory nerve (nerve trunk) or motor nerve branch or intramuscular motor branch.[13] The targeted nerve is stimulated via the probe tip using an externally connected nerve stimulator at values of <1 mA with ultrasound guidance to verify target selection.[10,11] The probe rapidly cools to between −60°C and −88°C, depending on the cryogen boiling point, owing to the rapid expansion of a gaseous cryogen.[8,10,14] An ice ball forms around the probe tip because of freezing of the surrounding liquid in the tissues.[10] Cryoneurolysis causes second-degree nerve injury (axonotmesis) resulting in Wallerian degeneration while keeping the epineurium and perineurium intact, allowing for length-dependent axonal regeneration.[10,14,15] Previous case reports have reported improvements in spasticity after cryoneurolysis may be maintained through 17 mos.[8,10]

Cryoneurolysis can treat both motor and mixed motor sensory nerves in patients with spasticity.[13] The choice of nerve targeting is first established through diagnostic anesthetic nerve blocks, which outline the individual muscle contribution to the spastic deformation around a joint.[16,17] The diagnostic anesthetic nerve block algorithm allows targeting of the motor and sensory components that cause spasticity and associated pain. Here, we prospectively evaluated the clinical, functional, and analgesic outcomes of the management of upper limb spasticity either refractory to or plateaued on botulinum therapies in participants treated with percutaneous ultrasound-guided cryoneurolysis.

METHODS

Study Design

This repeated-measures observational pilot study (NCT04670783) performed at a single academic institution included participants who underwent cryoneurolysis of upper limb nerves. Participants were followed for up to 1 yr. The study was approved by the local research ethics boards (Vancouver Island Health and UBC Clinical Research Ethic Boards). Written informed consent was obtained for each participant. The study was conducted between October 8, 2020, and May 30, 2023, and conforms to all strengthening the reporting of observational studies in epidemiology guidelines and reports the required information accordingly (Supplementary Checklist, http://links.lww.com/PHM/C722).

Participants

Adult participants were eligible if they had upper limb spasticity causing a functional impairment (e.g., less than desired range of motion [ROM] or movement to perform a task) in the shoulder, elbow, wrist, or fingers/thumb; plateaued in outcomes after standard spasticity treatments (e.g., botulinum toxin, physiotherapy, bracing, and oral medications); and undergone clinical examination of the upper limb (including a maximal passive ROM [V1] and fast catch) that demonstrated possibility of further ROM after an ultrasound- and e-stimulation-guided, diagnostic anesthetic nerve block with 1–2 ml of 2% lidocaine to each nerve target to determine if the muscle had reducible spasticity.[17] Participants who had received a previous botulinum injection were considered eligible if they had not received an injection in the past 4 mos. Participants were excluded if they could not attend the treatment follow-up schedule or received a previous neurolytic procedure to the nerve (e.g., phenol or cryoneurolysis) in the past 2 yrs. Participants were recruited at a multidisciplinary spasticity clinic over a 19-mo period. The diagnoses of the study population could include multiple causes of spasticity including multiple sclerosis, cerebral vascular accident (stroke), cerebral palsy, spinal cord injury, and traumatic brain injury. The protocol was designed as an outcome of the clinic's standard of care; participants were enrolled after electing to receive cryoneurolysis. Because this study is an observational study, no adjustments were made to the routine spasticity treatments participants received.

Procedures

All participants were assessed by a physiatrist or anesthesiologist. The diagnostic anesthetic nerve block to each nerve target screened participants to determine the muscle(s) most responsible for the spastic upper limb and to limit potential adverse events. Percutaneous cryoneurolysis was applied to the mixed motor sensory trunks, motor nerves, or intramuscular branches of the motor nerves. The individual muscles targeted included the pectoralis major and minor for the shoulder; the biceps brachii, brachialis, and brachioradialis for the elbow; the flexor carpi radialis, flexor carpi ulnaris, and pronator teres for the wrist; and the flexor digitorum superficialis, profundus, flexor pollicis longus, and intrinsic muscles for the fingers and thumb. Median and ulnar mixed motor sensory trunks were also treated. The suprascapular nerve was targeted in participants with severe pain localizing to the shoulder joint associated with spasticity. Two different cryoneurolysis devices were used (iovera°, Pacira BioSciences, Inc and Lloyd SL 2000, Neurostat Spembly Medical), and a previous published protocol was followed.[8,13] Surface anatomy was used to place the ultrasound probe, which visualized the targeted nerve. A small amount of lidocaine (<1 ml of 1% lidocaine to avoid diffusion) was used to raise the skin wheal, then a 16-gauge angiocatheter was inserted to protect the skin and assist in guiding the cryoprobe. Cryoneurolysis lesions were performed on the targeted branch using an in-plane ultrasound technique, with e-stimulation confirming the targeted region (Supplemental Digital Content, Video 1, http://links.lww.com/PHM/C723). Vascular structures were avoided in part by using the ultrasound color Doppler setting. For the iovera° device, a 106-second freeze-thaw cycle was performed for 1–3 lesions per targeted nerve. For the Lloyd SL 2000 device, one or two 3.5-min cycles were performed. Cryoneurolysis is believed to occur at −88°C for the iovera° device (nitrous oxide) and −60°C for the Lloyd SL 2000 device (carbon dioxide gas).[8,14] If a participant experienced a burning pain at the end of the cycles, an additional cycle was offered until pain resolved. In most cases, a simple bandage was applied to stop any skin bleeding. Participants were reassessed immediately after cryoneurolysis to adjust the management strategy, goals, and need for cryoneurolysis to any other muscles.

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Outcomes

The primary outcome was to determine improvement in active and passive ROM, spasticity, and limb function, with each participant serving as their own baseline control. Outcomes were measured at baseline and 1, 3, 6, 9, and 12 mos within a 45-day window (Supplemental Digital Content, Supplemental Fig. 1, http://links.lww.com/PHM/C724). Age and sex assigned at birth were captured at baseline. ROM was evaluated as active ROM (AROM) and V1 measured using a goniometer during flexion, abduction, and external rotation for the shoulder and extension for the elbow and wrist. Spasticity in the shoulder, elbow, and wrist was evaluated using the modified Ashworth scale (MAS; scale grade range, 0–4, where 0 = "no increase in muscle tone" and 4 = "affected part(s) rigid in flexion or extension"; 1+ was converted to 1.5 for numerical calculation).[18] Grip strength was assessed using a Jamar dynamometer in kg. Manual dexterity was assayed using the box and block test (scored as the number of blocks moved from one compartment to another in 1 min).[19] Resting hand position was evaluated and scored on the Keenan scale (score range, 1–5, where 1 = "minimum deformity" and 5 = "severe clenched fist"),[20] while hand function was assessed using the House Functional Scale (score range, 0–8, where 0 = "does not use" and 8 = "spontaneous use, complete").[21] The House classification (range, type I ["minimum deformity"] to type IV ["maximum deformity"]) assessed thumb deformity.[21] Video capture of each participant was performed at baseline and each follow-up visit and was securely stored. Upper limb disability was assessed using the self-administered qualitative Disabilities of the Arm, Shoulder, and Hand (DASH) questionnaire comprising 30 questions (item score range, 0–5, where 0 = "no difficulty completing task" and 5 = "unable to perform task"; total score range, 0–100, where 0 = "no disability" and 100 = "most severe disability").

The average pain experienced during the past 24 hrs was assessed before completing the questionnaire to establish a baseline and at follow-up visits using the Brief Pain Inventory Questionnaire (BPIQ), which has the following four pain items: "worst" pain in the last 24 hrs, "least" pain in the last 24 hrs, "average" pain, and pain "right now," each assessed on a scale ranging from 0 to 10, where 0 = "no pain" and 10 = "pain as bad as you can imagine."[22] Average pain severity was defined as mild (score 1–3), moderate (score 4–6), and severe (score 7–10) as measured by the BPIQ. Participant satisfaction was measured with the Goal Attainment Scale (GAS; range, −2 to +2; transformed into a T score [mean = 50]) with the baseline score of −1 for each goal. Per institutional and study protocols, any adverse effect reported required documentation and a treatment plan in the clinical medical chart, study records, and research database (Research Electronic Data Capture). Safety data were previously published as part of 3 ongoing studies of cryoneurolysis for spasticity.[13]

Statistical Analysis

A study size of 50 was targeted to provide sufficient statistical analysis power (>0.8). All measurements and testing were performed by the research team and not the treating physicians. The numbers of participants are reported at each time point to indicate how many participants were lost to follow-up. Change from baseline included only participants with baseline measurements and data at the specified time point. To confirm improvements in outcomes that were statistically significant, a paired *t* test on the difference of proportions (pretreatment vs posttreatment) was performed. Repeated-measures or linear mixed (multilevel) models of change were used to analyze within-person trajectories in all key outcomes for the repeated-measures data that spanned the 12-mo study period, with each individual serving as their own baseline control. A Wilcoxon signed rank test was used to analyze changes from baseline in V1, AROM, and MAS score after cryoneurolysis. Skewed distributions were observed within all MAS and some of the V1 outcomes at baseline or follow-up visits, and Shapiro-Wilk test was used to confirm the nonnormality attribute. Local polynomial regression models were included in a post hoc analysis to evaluate the longitudinal trends among participants with different baseline pain levels. All analyses were based on a one-sided test at a significance level of 0.05. Confidence intervals were reported for outcomes that had more than two participants where the change from baseline was not 0 and when all improvements were not identical (minimum = maximum).

RESULTS

Participant Disposition and Baseline Characteristics

Of 63 total participants enrolled, four were sent for surgery and not included in the study. A total of 59 participants received cryoneurolysis, and during the follow-up period, four participants died from unrelated causes. After treatment, one participant withdrew from the study after the 1-mo follow-up, three participants withdrew because of transportation difficulties within 6 mos, and 1 participant withdrew after 7 mos. No participants discontinued because of adverse events or lack of improvement. In the 12 mos after cryoneurolysis, three of the enrolled participants received botulinum toxin. All three participants had maintained a significant improvement in V1 at the time of injection and at the postinjection follow-up, none of these participants showed a significant change in V1 in response to botulinum toxin injection.

Of 59 participants who received cryoneurolysis, 47 received cryoneurolysis for the shoulder, 33 received cryoneurolysis for the elbow, 18 received cryoneurolysis for the wrist, and 29 received cryoneurolysis for the fingers/thumb; a total of 182 nerves were treated. More than half of participants received cryoneurolysis in >1 region (n = 32), and nine participants received follow-up treatment in additional locations after initial treatment (Supplemental Digital Content, Supplemental Table 1, http://links.lww.com/PHM/C725). The mean (SD) age of participants was 59.5 (16.2) yrs, and 52.5% of participants were female. As reported in a previous study, the underlying diagnoses were predominantly cerebrovascular accident (stroke; n = 49), spinal cord injury (n = 3), cerebral palsy (n = 2), multiple sclerosis (n = 2), and traumatic brain injury (n = 2).[13] Participant demographics and baseline characteristics are shown in Table 1.

Shoulder Cryoneurolysis Outcomes

Participants had significantly sustained improvements in V1 at every time point compared with baseline ($P \leq 0.001$

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for all) across shoulder abduction, external rotation, and flexion measures (Fig. 1; Supplemental Digital Content, Supplemental Table 2, http://links.lww.com/PHM/C725, and Videos 2–4, http://links.lww.com/PHM/C726, http://links.lww.com/PHM/C727, http://links.lww.com/PHM/C728). At 3 and 12 mos, AROM was significantly improved for abduction and flexion ($P \leq 0.03$ for all) with the largest improvement occurring at 12 mos for abduction and flexion (mean [SD] change from baseline, +9.7° [15.5°; 95% confidence interval (CI), 2.5°, 20.0°] and +8.1° [17.8°; 95% CI, 0°, 20.0°], respectively). Sustained significant improvements in MAS scores were observed across abduction, external rotation, and flexion measures compared with baseline ($P \leq 0.001$ for all); at 12 mos, most MAS scores were at or below the 1+ grade. The greatest improvements in MAS scores were observed at 12 mos for abduction, external rotation, and flexion (mean [SD] change from baseline, −2.0 [0.8; 95% CI, −2.5, 1.8]; −1.4 [1.2; 95% CI, −2.3, −1.3]; and −1.7 [0.9; 95% CI, −2.3, −1.5], respectively). For all measures, the average trend line for V1 slightly increased while the average trend line for MAS score slightly decreased over time (Fig. 2A); observed trends were similar for both cryoneurolysis devices used (Supplemental Digital Content, Supplemental Fig. 2, http://links.lww.com/PHM/C724). At 12 mos, 46% (12/26), 48% (11/23), and 48% (12/25) of participants experienced >20° improvements in V1 abduction, rotation, and flexion, respectively (Fig. 2B).

TABLE 1. Baseline demographics and characteristics

	Value	*n*
Age	59.5 (16.2)	59
Sex, %		
Female	52.5	31
Male	45.8	27
NA	1.7	1
Shoulder		47
V1, °		
Abduction	94.6 (18.6)	47
External rotation	30.0 (24.3)	46
Flexion	94.7 (20.0)	46
AROM,°		
Abduction	58.1 (18.1)	36
External rotation	14.7 (26.6)	16
Flexion	55.7 (20.9)	34
MAS score		
Abduction	3.4 (0.9)	44
External rotation	2.9 (1.1)	42
Flexion	3.3 (0.8)	43
Elbow		33
Extension V1, °	−39.6 (29.8)	33
Extension AROM, °	−59.3 (27.4)	21
Extension MAS	3.1 (1.1)	33
Wrist		18
Extension V1, °	37.5 (37.7)	17
Extension AROM, °	22.9 (27.1)	7
Extension MAS	3.0 (1.0)	17
Fingers/Thumb		29
House Functional Scale score	0.68 (1.5)	25
House classification		11
Type I	NA	9
Type II	NA	7
Type III	NA	1
Type IV	NA	7
Grip strength		25
Zero	0 (0)	18
Nonzero	3.6 (4.6)	7
Box and block test, blocks/min	2.6 (4.0)	21
Keenan scale score		21
1	NA	3
2	NA	5
3	NA	0
4	NA	2
5	NA	11
GAS score	36.8 (1.4)	50
BPIQ score		33
Average pain	4.0 (2.1)	33
Pain right now	2.3 (2.7)	33
Least pain in the last 24 hrs	1.8 (1.9)	33
Worst pain in the last 24 hrs	5.6 (2.5)	33
DASH score, final	64.3 (21.8)	56

Values are the mean (SD) unless indicated otherwise.

Elbow Cryoneurolysis Outcomes

Participants had improvements in elbow extension V1, AROM, and MAS score over the follow-up period (Supplemental Digital Content, Videos 2–4, http://links.lww.com/PHM/C726, http://links.lww.com/PHM/C727, http://links.lww.com/PHM/C728). At 1 and 12 mos after treatment, V1 significantly improved (mean [SD] change from baseline, +8.6° [15.2°; 95% CI, 2.5°, 22.5°]; $P = 0.009$ and +12.9° [18.2°; 95% CI, 2.5°, 35.0°]; $P = 0.007$, respectively) (Fig. 3A; Supplemental Digital Content, Supplemental Table 3, http://links.lww.com/PHM/C725). Of note, 3 participants (9%) had almost full elbow extension (≤15°) at baseline and could not exhibit further improvements. There was a significant increase in AROM at 12 mos (mean [SD] change from baseline, +21.3° [27.0°; 95% CI, 7.5°, 47.5°]; $P = 0.007$). Compared with baseline, elbow extension MAS score was significantly improved at all time points after cryoneurolysis ($P \leq 0.008$ for all).

Wrist Cryoneurolysis Outcomes

V1 and AROM measurements nonsignificantly increased at 1, 3, 6, and 12 mos compared with baseline (Fig. 3B; Supplemental Digital Content, Supplemental Table 4, http://links.lww.com/PHM/C725). There was a significant decrease in wrist extension MAS score at 3 and 12 mos (mean [SD] change from baseline, −1.0 [1.1; 95% CI, −1.8, −0.5]; $P = 0.01$ and −1.3 [0.8; 95% CI, −2.0, −1.3]; $P = 0.03$, respectively). Among all individuals, mean change from baseline showed improvement across all 3 measures (Supplemental Digital Content, Supplemental Fig. 3, http://links.lww.com/PHM/C724).

Fingers and Thumb Cryoneurolysis Outcomes

Hand function improved significantly at 3 mos (mean [SD] change from baseline in House Functional Scale score, 0.6 [1.2; 95% CI, 1.0, 4.0]; $P = 0.03$) but not at 12 mos for

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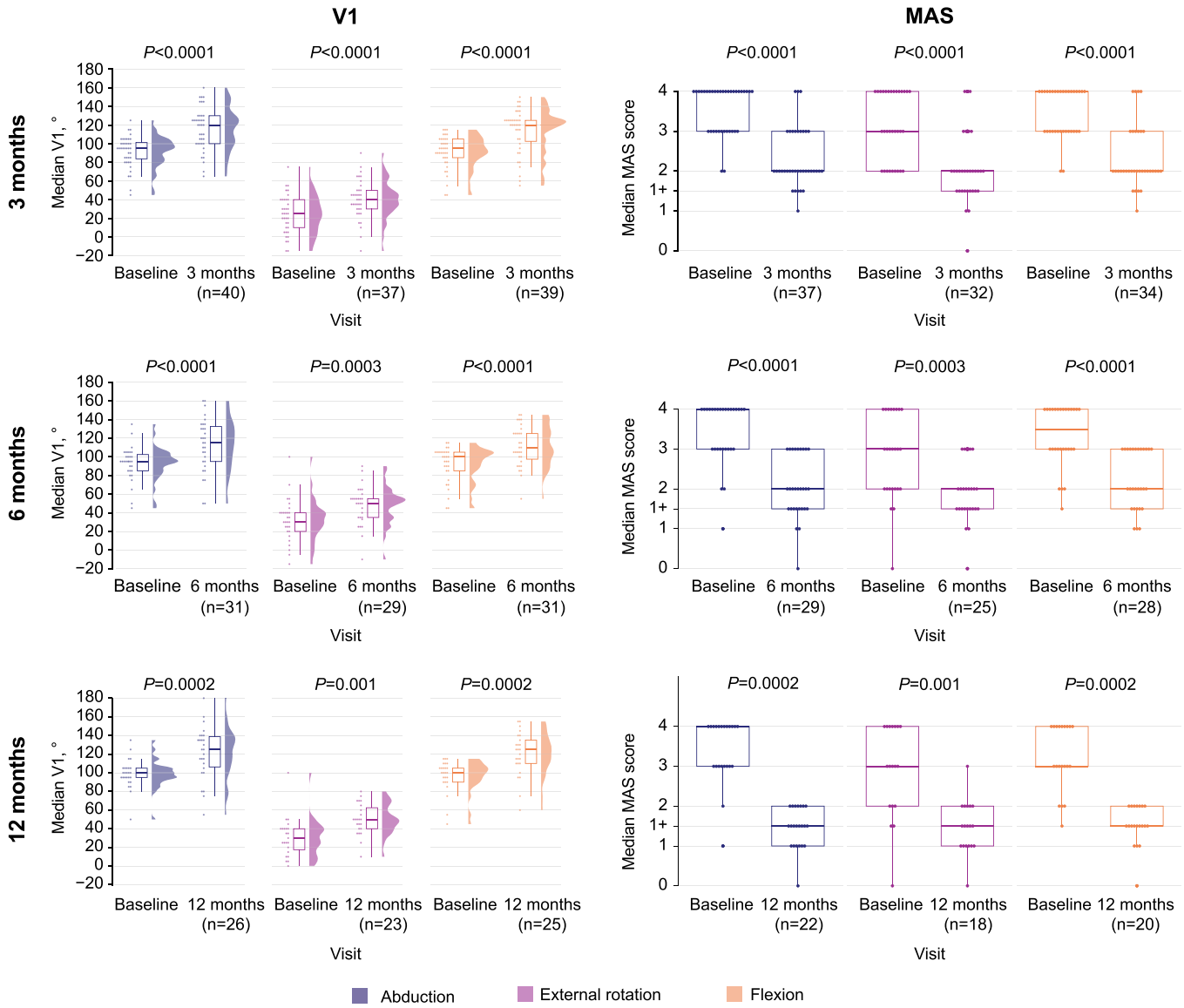

FIGURE 1. Maximal passive range of motion (left panels) and MAS scores (right panels) for shoulder abduction, external rotation, and flexion at baseline and at 3, 6, and 12 mos. Boxplots show the IQR, lines show the median, error bars show median ± 1.5 × IQR, data points represent data for individual participants, and shading represents a corresponding violin plot.

participants who received fingers/thumb cryoneurolysis ($n = 29$; Supplemental Digital Content, Supplemental Table 5, http://links.lww.com/PHM/C725). However, 20 participants had a nonfunctional hand (House Functional Scale score of 0) at baseline and were not expected to improve. At 12 mos, ordinal improvements in thumb position were observed in 2 participants (baseline House classification type IV reaching type II and baseline House classification type II reaching type I; Supplemental Digital Content). For participants with no grip strength at baseline in the affected hand, there were slight increases in grip strength over time; changes stratified by participants with and without measurable grip strength at baseline are shown in the Supplemental Digital Content (Supplemental Fig. 4A, http://links.lww.com/PHM/C724). At 12 mos, more blocks were picked up for the 2 participants who had box and block test scores (mean [SD] change from baseline, 15 [19.8] blocks/min) but this was not significant (Supplemental Digital Content, Supplemental Fig. 4B, http://links.lww.com/PHM/C724). Significant improvement in resting hand position was observed at 3 mos (improved ratio change from baseline in Keenan scale score, 0.3 [95% CI, 0.02, 0.8]; $P = 0.03$) (Supplemental Digital Content, Supplemental Table 5, http://links.lww.com/PHM/C725).

Pain, Upper Limb Disability, and Participant Satisfaction Outcomes

Across all study participants, average pain was significantly reduced from baseline at 1, 3, 6, and 12 mos ($P \leq 0.04$ for all), with pain "right now" significantly reduced at 6 mos (mean [SD] change from baseline, −1.1 [2.1; 95% CI −7.5, −3.5]; $P = 0.02$) (Fig. 4A; Table 2). A decreasing, but nonsignificant, trend in average pain was observed across pain severity at all time points (Fig. 4B). Total DASH scores were

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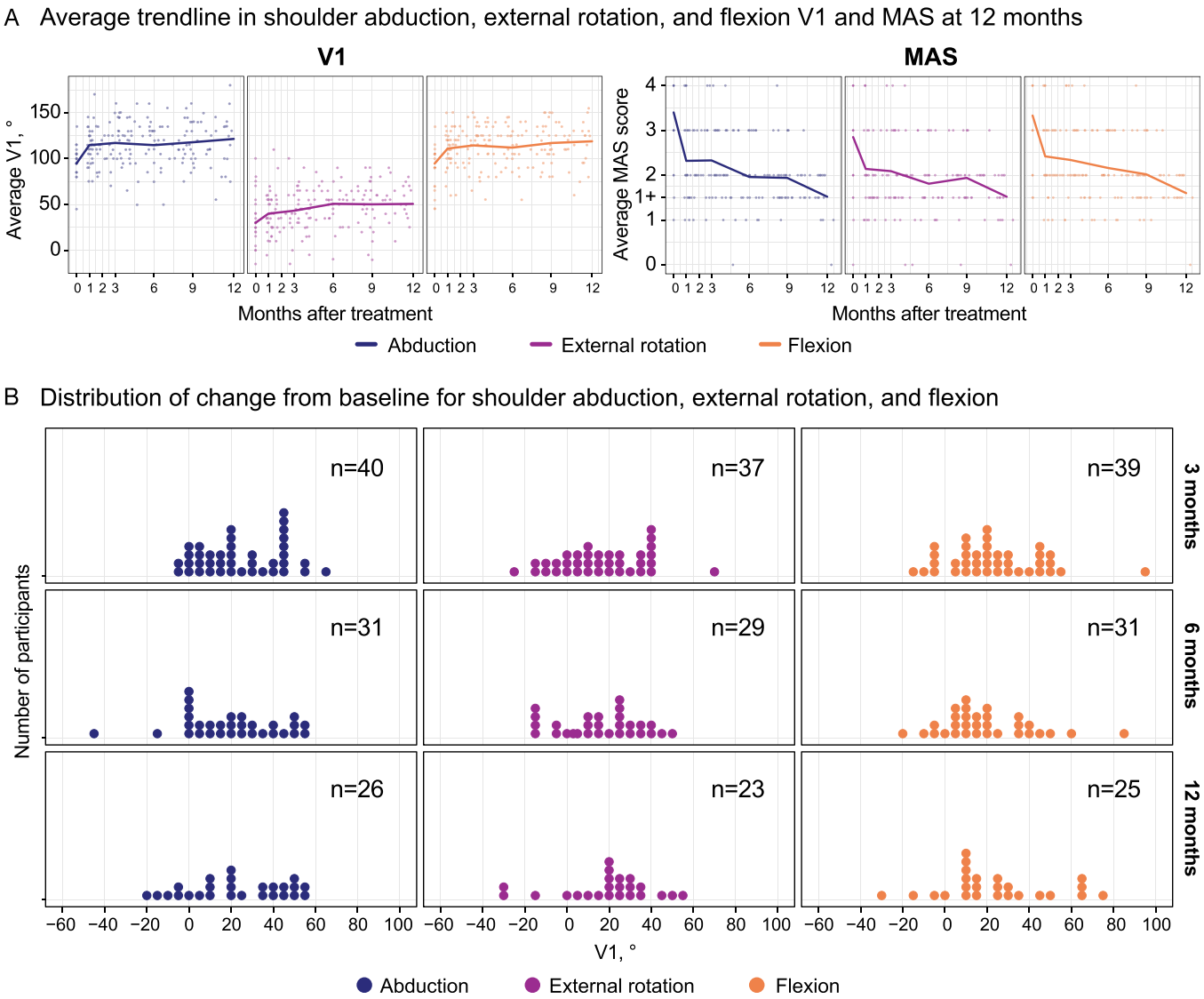

FIGURE 2. A, Trend lines across individual participant data points showing average V1 (left panels, in °) and MAS scores (right panels) for shoulder abduction, external rotation, and flexion over the follow-up period. B, Distribution of change for individual participants from baseline for shoulder abduction, external rotation, and flexion V1 at 3, 6, and 12 mos. The observation of negative V1 change from baseline for participants at 3 mos was partly attributable to four participants, 3 who died before study completion (and had negative results) and 1 who had severe pain due to unrelated complex regional pain syndrome. At 6 mos, participants with negative changes in V1 relative to baseline included 1 who had a humerus fracture, 3 who had severe pain (complex regional pain syndrome, impingement syndrome, capsulitis), and 1 who was referred for surgery at 6 mos. The remainder of participants exhibiting negative changes in V1 at 3 or 6 mos demonstrated decreased spastic tone (MAS) and improvement of V1 in other shoulder directions.

significantly reduced compared with baseline at 1, 3, 6, and 9 mos ($P < 0.04$). Among participants with data at 12 mos, DASH total score was nonsignificantly reduced (mean [SD] change from baseline, −9.3 [17.8; 95% CI, −29.0, 6.5]; $P = 0.08$) (Fig. 4C). There was a mean (SD) increase of 11.9 (9.8) points in GAS score at 12 mos compared with baseline (Supplemental Digital Content, Supplemental Table 6, http://links.lww.com/PHM/C725), with >33% of participants having improvement in their most important (primary) goal.

DISCUSSION

Percutaneous cryoneurolysis targeting multiple upper limb nerves resulted in significant improvements in shoulder, elbow, and wrist spasticity and hand function from baseline. Participants with shoulder spasticity had improved shoulder V1, abduction and flexion AROM, and spastic tone out to 12 mos. In participants who received elbow flexor muscle cryoneurolysis, significant improvements in AROM, V1, and MAS scores were observed at 12 mos, and in participants who received wrist flexor muscle cryoneurolysis, MAS score was significantly improved at 12 mos. Improvements in spasticity were clinically meaningful for the shoulder and elbow at all time points and for the wrist at 3 and 12 mos per the previously defined MAS score minimal clinically important difference (MCID) threshold after stroke.[23] Hand function significantly improved at 3 mos, with numerical improvements at 12 mos although the sample size at 12 mos was too low to

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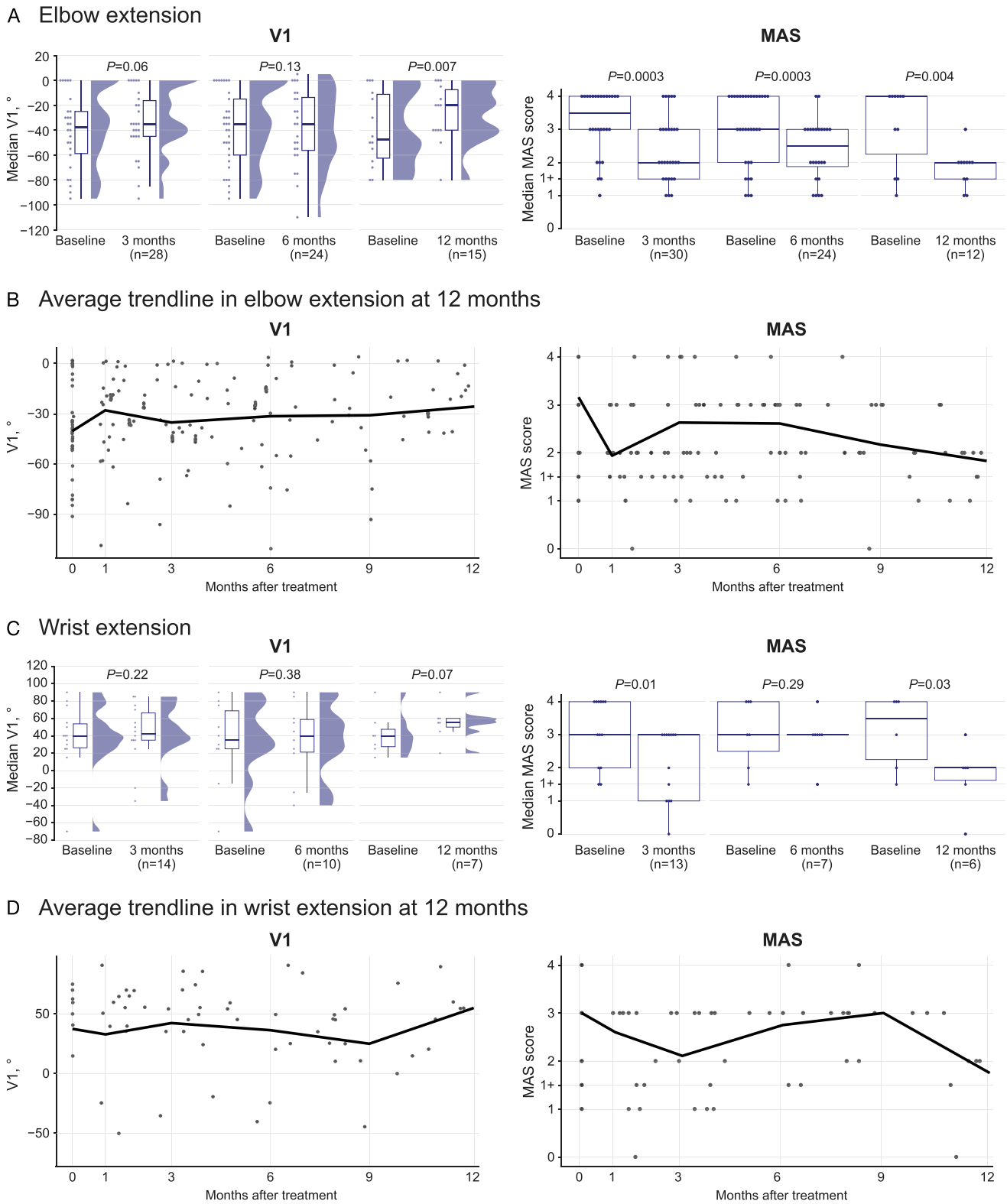

FIGURE 3. Outcomes after elbow and wrist cryoneurolysis over time. A, Maximal passive range of motion (left panels) and MAS (right panels) scores for elbow extension at baseline and 3, 6, and 12 mos. Boxplots show the IQR, lines show the median, error bars show the median ± 1.5 × IQR, data points represent data for individual participants, and shading represents a corresponding violin plot. B, Trend lines across individual participant data points showing average V1 (left panel) and MAS (right panel) for elbow extension to 12 mos. C, Maximal passive ROM (V1; left panels) and MAS (right panels) scores for wrist extension at baseline and 3, 6, and 12 mos. Boxplots show the interquartile range, lines show the median, error bars show the median ± 1.5 × IQR, data points represent data for individual participants, and shading represents a corresponding violin plot. D, Trend lines across individual participant data points showing average V1 (left panel) and MAS (right panel) for wrist extension to 12 mos.

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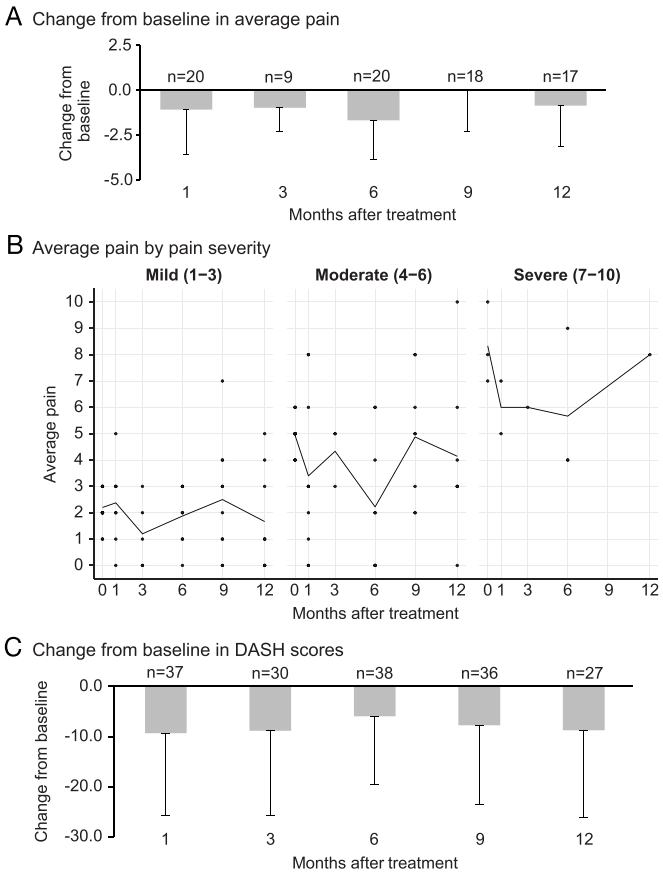

FIGURE 4. A, Change from baseline in average pain. B, Trend lines across individual participant data points showing average pain by severity (as measured by the Brief Pain Inventory Questionnaire) for all participants (data points represent individual participants; not all participants are visible because of overlap of scores for individual participants. C, Change from baseline in final DASH scores over time for all participants regardless of treatment location. Error bars show the SD.

speculate on the lack of statistical significance at this time point. While changes in box and block scores from baseline were not significant at 3, 6, and 12 mos, they surpassed the MCID defined for change in box and block scores among patients who experienced a stroke.[24] Cryoneurolysis resulted in progressive pain reductions, with significant reductions in

TABLE 2. Change from baseline in BPIQ scores at 1, 3, 6, 9, and 12 mos

	Average Pain	Worst Pain in Last 24 hrs	Least Pain in Last 24 hrs	Pain Right Now
1 mo	*n* = 20	*n* = 20	*n* = 20	*n* = 20
Mean (SD)	−1.1 (2.5)	−0.9 (2.8)	−0.3 (2.2)	−0.5 (2.7)
95% CI	−8.0, −4.0	−7.5, −3.0	−6.5, −3.0	−8.0, −3.5
P value	0.02	0.08	0.32	0.23
3 mos	*n* = 9	*n* = 9	*n* = 9	*n* = 9
Mean (SD)	−1.0 (1.3)	−0.9 (3.5)	0.1 (2.7)	0.6 (2.9)
95% CI	−7.0, −3.0	−8.0, −1.5	−7.0, −0.5	−6.5, 0
P value	0.04	0.22	0.53	0.83
6 mos	*n* = 20	*n* = 20	*n* = 20	*n* = 20
Mean (SD)	−1.7 (2.2)	−1.2 (3.2)	−0.6 (1.6)	−1.1 (2.1)
95% CI	−8.0, −4.0	−8.0, −3.5	−6.5, −4.0	−7.5, −3.5
P value	0.003	0.06	0.09	0.02
9 mos	*n* = 18	*n* = 18	*n* = 18	*n* = 18
Mean (SD)	0 (2.3)	−0.7 (2.4)	0.06 (2.5)	1.4 (2.8)
95% CI	−5.0, −2.5	−6.5, −3.0	−5.0, −2.5	−4.0, −0.5
P value	0.43	0.11	0.46	0.97
12 mos	*n* = 17	*n* = 17	*n* = 17	*n* = 17
Mean (SD)	−0.9 (2.3)	−0.6 (2.4)	0 (1.9)	0.3 (2.1)
95% CI	−7.0, −3.0	−6.5, −3.0	−6.0, −2.5	−5.0, −2.0
P value	0.04	0.17	0.54	0.62

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average pain at 12 mos. One-third of participants met their primary goals. Overall, cryoneurolysis improved spastic tone, active and passive ROM, and participant satisfaction and reduced pain out to 12 mos. Notably, the study population was refractory to other treatments, providing evidence that future improvements in ROM, function, and pain are possible after a plateau.

Previous studies and case reports investigating percutaneous cryoneurolysis reported pain relief associated with knee osteoarthritis, glenohumeral osteoarthritis, total knee arthroplasty surgery, and neuralgia.[25–28] In one case report, preoperative cryoneurolysis provided postoperative analgesia resulting in low pain scores (<2) using a numerical rating scale (range, 0–10) after shoulder cuff repair and total knee arthroplasty.[29] A patient with glenohumeral osteoarthritis had clinically important improved ROM, which was sustained 7 mos after cryoneurolysis treatment, in another case report.[28] A previous case series investigated cryoneurolysis treatment among participants with spasticity who had received botulinum injections; one participant had increased elbow ROM that persisted 17 mos after cryoneurolysis treatment.[8] Previously published safety results including participants from the current study indicated that cyroneurolysis is well tolerated in patients with spasticity with minimal and manageable side effects.[13] The results from the current study support and expand upon previous results of reduced pain and increased ROM after cryoneurolysis in participants with upper limb spasticity, demonstrating that these improvements persist to 12 mos.[8,28,29]

This study has numerous strengths including examining a large number of participants with spasticity using a wide array of subjective and objective outcomes after cryoneurolysis treatment. Notably, this study is one of three occurring at the same institution assessing the impact of cryoneurolysis on spasticity, associated pain, and safety in upper and lower limbs to provide an all-encompassing view of cryoneurolysis effects for patients with spasticity.[13] The study participants represented a wide range of ages and diagnoses, adding to the value. Additionally, the observational, prospective design permitted close follow-up of each participant to monitor outcomes and side effects with participants serving as their own control, permitting direct comparison to baseline. Not all participants respond to cryoneurolysis for each outcome measure, nor experience the same duration of effect[29]; thus, examining individual responses allows for data normalization to individual participants. Finally, follow-up of participants out to 12 mos helps establish the longevity of cryoneurolysis to treat spasticity in the upper limb.

This study also has limitations. Multicenter studies are needed to establish generalizability of the results, given this was a single center study. Additionally, typical of surgical studies, data collectors were not blinded, which could lead to ascertainment bias. Some outcome measures had a limited number of participants, not all participants completed every follow-up, and time points fell within a 45-day window of follow-up (i.e., follow-up did not occur at the same times for each participant). It is also possible that outcomes could be affected by receiving cryoneurolysis in different areas (i.e., if a participant received shoulder cryoneurolysis and follow-up elbow cryoneurolysis, shoulder cryoneurolysis may affect the elbow outcomes). There is a need for future head-to-head studies with other treatments used on- or off-label for spasticity (e.g., phenol) to determine the benefits of each treatment.

Overall, this repeated-measures pilot study suggests that percutaneous cryoneurolysis can provide long-lasting changes in spasticity measurements and function for the shoulder, elbow, wrist, and fingers/thumb while reducing pain and improving participant satisfaction, supporting future investigation of treating spasticity with cryoneurolysis.

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